

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

This observing program's purpose is to calculate the rotation period of a given asteroid, through measuring periodic changes in its apparent brightness by means of a lightcurve. This program is interesting as we would not only be able to observe a low-magnitude small body, but also be able to understand its dimensions and shape.

We have a list of potential targets of apparent magnitudes between 13-15, though the presumed main-target is 155 Scylla - which is a main-belt asteroid. We would be observing this/these targets through the Kuiper 61" telescope with the Mont4k CCD, taking 60-second exposures with an I-band filter attached.

Telescope Requested (0.5 points): *Kuiper 61"*

Instrument (0.5 points): *Mont4K, with I-band filter*

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

Bright, with the presumption that the asteroid isn't too close to the moon - we are measuring relative brightness, while moonshine may affect our measurement, it is not a glaring issue unless the Mont4k is completely unable to detect the asteroid.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Object Name	Right Ascension	Declination	Apparent V magnitude	Object Type	Visibility
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155 Scylla	00 51 54.10	-04 19 27.3	14.167	Asteroid	September through November
170 Maria	22 50 43.81	+13 26 04.4	13.209	Asteroid	September through October
188 Menippe	03 02 24.26	+24 23 46.2	13.391	Asteroid	September through November
167 Urda	01 36 58.47	+07 56 15.8	13.302	Asteroid	September through November
248 Lameia	02 33 32.76	+18 50 51.8	14.063	Asteroid	September through November
295 Theresia	01 39 24.80	+14 59 13.3	13.162	Asteroid	September through November
2 Pallas	01 19 37.37	-14 59 29.9	8.265	Asteroid	September through November
236 Honoria	23 05 13.51	-04 08 34.1	11.4	Asteroid	September through November
71 Niobe	23 12 31.07	+22 53 29.8	11.9	Asteroid	September through November
695 Bella	23 31 20.56	+24 18 41.0	12.0	Asteroid	September through November
321 Florentina	14 35 10.12	-15 31 53.2	15.982	Asteroid	September through November

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

For 155 Scylla - which I will be basing the program characteristics off of - the program will last ~12 hours (1.5x the measured rotation period: ~7.95 hours), with individual exposures lasting around 60 seconds, and with an SNR of at least 150.

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How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (**2 points**)

The asteroid is presumed to be elongated to a point where there are differences in apparent brightness, as the area of light reflecting off of its surface changes as it rotates. Our relative brightness measurements, as such, should change in a repeated fashion – to the point where if it were plotted on a scatterplot (time vs relative flux) it would produce a sinusoidal curve. We then determine the rotation period by figuring the time it takes for the initial highest/lowest relative brightness measurement to return.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (**0.5 points**)

In the event that we cannot observe, we would attain lightcurve data from online archival data repositories from the Planetary Data System website. We would take data from any of our potential targets, and process the data as we otherwise would from there.

<https://sbn.psi.edu/pds/resource/lc.html>

<https://iopscience.iop.org/article/10.1088/1538-3873/ae7cb6/pdf>